

Exercise no. 2

OMIM and PubMed

Part A - Inherited diseases Genes - OMIM

In this exercise, you will choose a human disease and find the Genbank accession numbers and sequences of some genes which are thought to affect it.

1. Go to OMIM database and perform a search for a human genetic disease you are interested in. Some possibilities include: Gaucher's, leukemia, sickle cell anemia.
Note the number of entries.

2. Save the first page of the result and select two entries in the results which you will use to find related nucleotide sequences in Genbank.

3. Display the chosen nucleotide sequence in FASTA format and save it.

Part B- Medical articles - PubMed

In this question, you will search for articles on your chosen disease and restrict your search in various ways.

1. Go to PubMed database and perform a search for a same human genetic disease you used in OMIM. Write down the number of articles you get.

2. Perform the same search, only for the articles that appeared exactly during the past one year.
Note the number of entries.

3. Save the abstracts of the first five search results, sorted by Author. Write down the name of the first author in the 5 articles you get.

4. Perform the search for the same human disease, for articles authored by either Smith or Jones. Write down the number of articles you get. Save the abstracts of the first five search results, sorted by Author. Write down the number of articles you get.

5. Save the summary description of the first 10 search results, sorted by Journal. Write down the Journal names that are listed.